

UNIVERSITY INSTITUTE OF ENGINEERING

Department Of Computer Science & Engineering

Bachelor of Computer Science ^{6th} Semester

Full Stack Development – II

(23CSH-309/23ITH-309)

Unit No. 3

Chapter No. 1

Lecture No. 1

Topic : Security Filter Chain

Er. Tushar Sood (E18319)

Technical Trainer(CSE)

Learning Objectives

1. Understand the concept of Security Filter Chain
2. Explain how Spring Security processes HTTP requests
3. Understand the role of filters in authentication and authorization
4. Explain the importance of filter order
5. Understand how Security Filter Chain improves application security

Security Filter Chain in Spring Security

- Spring Security protects applications by applying a series of security filters to every incoming HTTP request.
- These filters are organized into a Security Filter Chain, which determines how requests are authenticated, authorized, and processed before reaching application controllers.
- The Security Filter Chain is the core mechanism through which Spring Security enforces application security.

What is Spring Security?

- Spring Security is a powerful framework that provides authentication, authorization, and protection against common security threats.
- It integrates seamlessly with Spring Boot applications and follows a filter-based architecture.
- Spring Security is highly customizable and suitable for enterprise-grade applications.

What is a Security Filter?

- A security filter is a component that intercepts HTTP requests and responses.
- Filters examine, modify, or block requests based on defined security rules.
- Each filter performs a specific security task, such as authentication, authorization, or session management.



What is Security Filter Chain?

- The Security Filter Chain is an ordered collection of security filters executed for every incoming request.
- Each request passes through the chain before reaching the controller.
- If any filter fails (for example, authentication failure), the request is blocked immediately.



Purpose of Security Filter Chain

The Security Filter Chain exists to:

- Enforce authentication
- Apply authorization rules
- Prevent unauthorized access
- Protect against security attacks
- Maintain application integrity

It ensures that only valid and authorized requests reach business logic.

How Security Filter Chain Works

The request flow is as follows:

1. Client sends HTTP request
2. Request enters Security Filter Chain
3. Filters execute sequentially
4. Authentication and authorization are checked
5. Request reaches controller if permitted

This layered approach ensures strong security enforcement.



Filter Order Importance

- The order of filters in the chain is extremely important.
- Some filters must run before others to ensure proper security processing.
- For example, authentication filters must run before authorization filters.
- Incorrect order can lead to security vulnerabilities or runtime errors.



Common Filters in Security Filter Chain

Some commonly used filters include:

- Authentication Filter
- Authorization Filter
- Session Management Filter
- CSRF Filter
- Exception Translation Filter

Each filter has a specific and well-defined responsibility.



Authentication Filter

- Authentication filters verify user identity.
- They extract credentials (username, password, token) from requests and validate them.
- If authentication fails, the request is rejected immediately.



Authorization Filter

- Authorization filters determine what an authenticated user is allowed to do.
- They check user roles, authorities, and permissions before granting access.
- Authorization ensures principle of least privilege.



CSRF Filter

- CSRF (Cross-Site Request Forgery) filters protect applications from malicious request attacks.
- They validate tokens to ensure requests are legitimate.
- CSRF protection is essential for stateful web applications.



Exception Handling in Filter Chain

If an error occurs during filter execution:

- The exception is intercepted
- A proper error response is generated
- The request does not reach the controller

This ensures graceful handling of security errors.



Security Filter Chain Configuration

- In Spring Boot, the Security Filter Chain is configured using Java-based configuration.
- Developers define security rules declaratively, controlling authentication and authorization behavior.
- This approach provides flexibility and clarity.



Benefits of Security Filter Chain

Security Filter Chain provides:

- Centralized security management
- Strong request-level protection
- Modular and extensible security
- High scalability
- Consistent enforcement of security policies

It is essential for secure backend systems.



Security Filter Chain in REST APIs

In RESTful applications, the Security Filter Chain:

- Protects API endpoints
- Validates tokens (JWT, OAuth)
- Prevents unauthorized access
- Ensures stateless security

It plays a critical role in modern microservices.



Real-World Use Cases

Security Filter Chain is widely used in:

- Enterprise web applications
- REST APIs and microservices
- Banking and financial systems
- Cloud-based applications

It forms the backbone of application-level security.

Quiz/ FAQ's

1. Security Filter Chain is:

- A. A database component
- B. A collection of security filters
- C. A UI framework
- D. A REST controller

Answer: B

2. Filters are executed:

- A. Randomly
- B. In parallel
- C. In a defined order
- D. After controller execution

Answer: C

References/ Articles/ Videos

- Spring Security Reference Documentation

<https://spring.io/projects/spring-security>

- Spring Security Filter Chain

<https://docs.spring.io/spring-security/reference/servlet/architecture.html>

- Baeldung – Spring Security Filters

<https://www.baeldung.com/spring-security-filters>

- OWASP Security Concepts

<https://owasp.org>



Thank You

For queries

Email: tushar.e18319@cumail.in